Tangential acceleration

Consider an object moving along a circle of radius r. In a time Δt, the object travels an arc distance Δs as shown in Figure 2.47. The corresponding angle subtended is $\Delta\theta$

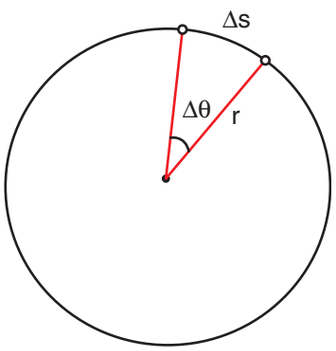


Figure 2.47 Circular motion

The Δs can be written in terms of $\Delta\theta$ as,

$$\Delta s = r\Delta\theta \qquad (2.35)$$

In a time Δt, we have

$$\frac{\Delta s}{\Delta t} = r\frac{\Delta\theta}{\Delta t} \qquad (2.36)$$

In the limit $\Delta t \to 0$, the above equation becomes

$$\frac{ds}{dt} = r\omega \qquad (2.37)$$

Here $\frac{ds}{dt}$ is linear speed (v) which is tangential to the circle and ω is angular speed. So equation (2.37) becomes

$$v = r\omega \qquad (2.38)$$

which gives the relation between linear speed and angular speed.

Note

The direction of linear velocity $\vec{v}$ is tangential to the circle, whereas the direction of angular velocity $\vec{\omega}$ is along the axis of rotation. The radius is also represented as a vector $\vec{r}$ directed radially from the centre of the circle.

Equation (2.38) is true only for circular motion. In general the relation between linear and angular velocity is given by

$$\vec{v} = \vec{\omega} \times \vec{r} \qquad (2.39)$$

For circular motion equation (2.39) reduces to equation (2.38) since $\vec{\omega}$ and $\vec{r}$ are perpendicular to each other.

Differentiating the equation (2.38) with respect to time, we get (since r is constant)

$$\frac{dv}{dt} = \frac{rd\omega}{dt} = r\alpha \qquad (2.40)$$

Here $\frac{dv}{dt}$ is the tangential acceleration and is denoted as a_t. $\frac{d\omega}{dt}$ is the angular acceleration α. Then eqn. (2.39) becomes

$$a_t = r\alpha \qquad (2.41)$$

The tangential acceleration a_t experienced by an object in circular motion is shown in Figure 2.48.

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